

		MIS 1	MIS 5e	MIS 7a-c	MIS 7e	MIS 9e	MIS 11c	MIS 13a	MIS 15a	MIS 15e	MIS 17	MIS 19c	MIS 21e	MIS 25	MIS 27	MIS 29	MIS 31	MIS 33	MIS 35	MIS 37	MIS 39	MIS 41	MIS 43	MIS 45	MIS 47	MIS 49
Benthic $\delta^{18}\text{O}$ (‰)	Lisiecki & Raymo (2005)	3.18	3.10	3.48	3.44	3.19	3.11	3.47	3.39	3.49	3.50	3.48	3.42	3.30	3.72	3.57	3.21	3.67	3.53	3.33	3.56	3.59	3.41	3.49	3.24	3.31
	Ahn et al. (2017)	3.22	3.06	3.48	3.37	3.15	3.15	3.61	3.39	3.46	3.61	3.44	3.40	3.28	3.67	3.64	3.26	3.65	3.47	3.29	3.60	3.46	3.29	3.49	3.25	3.30
	Zhou et al. (2026) - BIGSTACK_mixed	3.14	3.02	3.38	3.35	3.08	3.05	3.50	3.31	3.37	3.52	3.38	3.32	3.19	3.62	3.49	3.14	3.63	3.39	3.17	3.50	3.43	3.21	3.43	3.20	3.27
	Zhou et al. (2026) - BIGSTACK_mixedA	3.16	2.98	3.29	3.24	3.06	3.05	3.51	3.36	3.36	3.50	3.27	3.20	3.16	3.57	3.47	3.15	3.42	3.32	3.08	3.47	3.33	3.15	3.37	3.05	3.21
	Zhou et al. (2026) - BIGSTACK_mixedP	3.14	3.14	3.45	3.41	3.15	3.07	3.47	3.37	3.49	3.42	3.33	3.29	3.12	3.54	3.55	3.02	3.59	3.38	3.14	3.45	3.44	3.18	3.30	3.07	3.22
ΔSST (°C)	Clark et al. (2024) - Global Mean ΔSST stack	0.14	0.64	-0.48	-0.63	0.06	0.64	-1.17	-0.92	-0.50	-0.78	-0.71	-0.62	-0.57	-1.37	-0.38	0.52	-0.26	0.65	0.44	-0.30	0.53	1.14	0.29	1.06	1.08
	Clark et al. (2024) - North Atlantic ΔSST stack	0.37	1.78	-0.91	-0.78	1.06	1.28	-0.78	-1.00	-0.28	0.99	-0.87	-0.14	-0.70	-1.40	-1.79	2.27	-0.46	1.65	0.57	-0.74	1.31	2.69	-1.06	1.15	1.71
	Clark et al. (2024) - Tropical Atlantic ΔSST stack	0.31	1.13	0.45	-0.05	0.63	-0.34	-1.72	-0.60	-0.46	-0.59	-1.02	0.33	-0.96	-1.67	-0.05	0.69	0.41	1.23	0.14	0.90	2.32	1.15	2.80	2.34	2.42
	Clark et al. (2024) - South Atlantic ΔSST stack	0.63	1.86	-0.54	-0.21	1.84	1.93	-1.75	-1.57	-0.87	-0.66	-0.66	-0.13	1.63	-0.22	0.54	0.85	-1.20	1.43	1.39	0.72	1.36	0.90	1.38	2.77	1.60
	Clark et al. (2024) - North Pacific ΔSST stack	0.94	4.02	2.34	1.39	3.36	2.64	-0.24	0.91	1.26	1.21	1.19	1.62	0.62	0.16	0.57	2.11	-0.42	3.33	1.94	1.52	2.16	3.74	2.29	4.33	2.61
	Clark et al. (2024) - Tropical Pacific ΔSST stack	0.28	0.56	-0.41	-0.25	-0.09	-0.26	-0.82	-0.03	-0.16	0.10	-0.73	-0.25	0.03	-0.38	-0.24	0.54	0.18	0.26	0.07	-0.30	0.01	0.73	0.34	0.51	0.68
	Clark et al. (2024) - South Pacific ΔSST stack	0.67	2.35	0.19	-0.39	0.58	1.64	-0.28	-1.12	0.63	0.70	0.76	-0.04	0.51	-0.05	2.17	0.51	-0.08	-0.01	-0.30	1.13	1.60	1.56	1.81	1.71	1.70
	Clark et al. (2024) - Tropical ΔSST stack	0.27	0.80	0.01	-0.05	0.26	-0.09	-0.64	0.14	0.12	0.31	-0.27	0.21	0.15	-0.23	0.23	0.95	0.64	0.86	0.47	0.20	0.43	1.07	0.84	1.13	1.31
SST (°C)	Herbert et al. (2017)	27.01	27.66	26.95	27.18	27.30	27.53	26.94	27.13	27.13	27.28	27.15	27.45	27.62	26.85	27.03	27.52	27.08	27.73	27.92	26.91	27.60	28.41	27.57	28.12	28.12
Mean loess size (normalised)	Ding et al. (2002)	0.38	0.70	0.61	0.62	0.67	0.73	0.82	0.85	0.82	0.84	0.73	0.82	0.86	0.77	0.84	0.84	0.85	0.85	0.67	0.74	0.83	0.87	0.81	0.86	0.87
Mean size (μm)	Sun et al. (2021)	16.95	16.77	16.80	16.66	18.21	17.23	14.74	15.72	16.19	14.23	15.95	16.23	18.87	18.52	20.90	15.49	19.67	17.54	20.58	17.44	17.62	13.35	16.99	14.85	16.66
	Sun et al. (2019)	21.63	21.59	19.77	20.42	20.68	19.20	17.77	20.05	18.16	17.96	20.41	18.96	18.95	21.55	20.07	18.04	21.31	18.26	18.80	21.53	21.23	20.88	20.55	21.43	22.50
Inorganic loess carbonates $\delta^{13}\text{C}$ (‰)	Sun et al. (2019)	-6.26	-8.19	-6.31	-6.14	-6.11	-6.35	-6.35	-5.82	-6.46	-7.01	-6.24	-6.32	-5.35	-4.77	-6.53	-6.38	-4.89	-5.80	-6.26	-5.31	-4.85	-5.73	-6.25	-5.94	-4.81
MAP (mm/day)	Sun et al. (2019)	2.45	2.58	2.50	2.54	2.53	2.53	2.44	2.52	2.45	2.46	2.44	2.47	2.54	2.51	2.40	2.60	2.38	2.49	2.45	2.43	2.43	2.54	2.43	2.50	2.53
MAT (°C)	Sun et al. (2019)	12.07	12.18	11.30	11.87	11.94	12.23	10.99	11.61	11.41	11.22	10.64	11.15	11.70	11.69	10.96	11.74	11.18	11.22	11.31	10.99	10.77	11.42	10.99	11.77	11.60
	Zhao et al. (2021)	0.56	0.27	-1.80	-1.61	-1.50	-1.98	-1.00	-1.42	-0.58	-0.35	-0.84	-0.58	-0.57	-0.89	0.31	1.02	-1.04	-0.31	-0.34	-1.33	-0.09	-0.46	-0.89	-0.32	-1.08
	Wang et al. (2025)	nan	nan	nan	nan	10.54	11.73	10.89	11.97	13.02	11.69	11.96	12.35	15.07	11.84	9.14	12.43	10.56	10.40	11.15	8.52	8.92	10.58	9.53	10.25	12.13
AP (excl. Graminea and aquatic/semi aquatic taxa) (%)	Tzedakis et al. (2006)	99.48	96.76	95.81	96.70	95.50	100.00	95.78	96.90	95.18	89.08	83.27	93.20	95.67	54.63	72.77	95.00	84.98	85.84	90.38	75.88	76.64	91.89	nan	nan	nan
AP (excl. pinus) (%)	Donders et al. (2021)	80.73	92.78	95.95	85.02	91.46	81.20	93.46	92.37	88.17	65.73	82.96	87.50	88.36	67.88	74.79	93.43	87.20	97.01	98.52	96.62	93.21	90.52	nan	nan	nan
AP (%)	Zhao et al. (2020)	51.26	86.21	43.83	50.33	64.57	79.31	75.33	54.55	76.19	91.93	65.94	92.12	69.38	93.36	89.46	83.58	83.35	76.44	89.55	85.90	65.16	64.27	78.99	86.72	71.40
AP (%), excl. Quercus & Alnus	Torres et al. (2013)	nan	68.04	52.43	53.30	53.01	47.02	40.15	34.66	35.47	54.35	45.98	46.32	48.33	49.73	41.87	44.90	48.72	42.83	50.14	37.01	36.75	30.48	28.74	36.58	44.67
Si/Ti	Melles et al. (2012)	0.63	0.69	0.71	0.80	1.34	2.35	0.89	1.04	1.04	1.31	1.12	0.94	1.22	0.84	0.92	1.99	0.86	0.92	0.97	1.07	1.05	1.16	0.99	0.65	1.44
Biogenic Silicia (BioSi %)	Prokopenko et al. (2006)	nan	43.10	40.10	32.30	43.00	46.20	37.30	47.40	44.00	42.10	37.20	47.00	44.30	16.80	26.40	46.10	9.10	39.10	51.00	19.90	31.10	19.10	40.30	37.00	34.00
Log Ca/K	Sun et al. (2021)	4.10	4.81	3.89	3.69	3.71	4.58	4.09	4.24	4.46	4.33	4.28	4.28	4.03	2.49	3.72	4.61	3.44	4.35	4.98	3.52	4.27	5.75	nan	nan	nan